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[Paper 2] OLD VERSION

This page is the workspace for working groups and individuals engaged in pushing out Paper "2".

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1. Goals of Paper "2"

- (1) This is a comparison of dark matter-only simulations in a cosmological setup to investigate the agreements and discrepancies across different N-body calculation platforms in detail. This paper will complement the Flagship paper of the Collaboration (Paper "1").
- (2) Desired outcomes: snapshots of dark matter distribution at low redshift across all participating codes, comparison of simulations results in various dimensions

2. Timelines in 2014

- (1) Push out Paper "2" in **fall** of 2014
- (2) **Get Rockstar to run on multi-mass simulations (by end 08/2014)**
- (3) **Running of high-resolution simulations by end of 10/2014 the very latest, medium-res by end of 9/2014 the very latest**
- (4) **Transfer of snapshots to NERSC as they become available**
- (5) **Working group telecon mid September 2014.**

3. Overview of the Initial Conditions and Runtime Parameters

- (1) Dark matter only simulation of a $\sim 10^{11} M_{\text{sun}}$ halo with a quiescent merger history
- (2) MUSIC parameter file: [this link](#) (WG III's Workspace)
 - the same target halo and very similar to the parameter used in Paper "1", but,
 - the shape of the highest resolution region has been changed from a cuboid to an ellipsoid, therefore,
 - the .conf file of the IC used in Paper "1" (Nov. 20, 2012) is superseded by a newer file (Jan. 17, 2014)
 - participants will need to use MUSIC v1.5 (see the [v1.5 release note](#) in [MUSIC Google group](#))
- (3) Change levelmax=12 to 13 to increase particle mass resolution
- (4) Target halo mass measured in a test run at $z=0$ (4096^3 effective resolution, coarse refinement):

$$M_{\text{vir}} \sim 1.8 \times 10^{11} M_{\text{sun}} \text{ (see note 1), or } M_{200} \sim 1.7 \times 10^{11} M_{\text{sun}} \text{ (see note 2)}$$
- (5) **Force Resolution:** One AMR code should attempt to over-resolve the initial conditions (i.e. have more than 1 cell/particle initially) to agree with tree codes that typically start off with a PM with 2x the inter-particle resolution. See also the discussion in [Klypin+\(2013\)](#).
- (6) **Snapshots for Analysis:** We recommend total 201 timesteps: the first 21 timesteps equally spaced in redshift, $\Delta z = 0.5$, from $z=20$ to $z=10$, and the remaining 180 timesteps equally spaced in $\log(\text{expansion parameter})$, $\Delta \log(a) = |\log(1/11)/180|$, from $z=10$ to $z=0$. Plus redshift snapshots at $z = 6.0, 3.0, 2.0, 1.0, 0.5, 0.2, 0.0$ at the very least. The explicit list of outputs can be downloaded [here](#) (thanks to Junhwan Choi). **If a code group has already run with at least 100 outputs, it's ok to keep that run.**

[note 1] M_{vir} is defined as the mass enclosed in a sphere of mean density contrast of 360 with respect to the background density

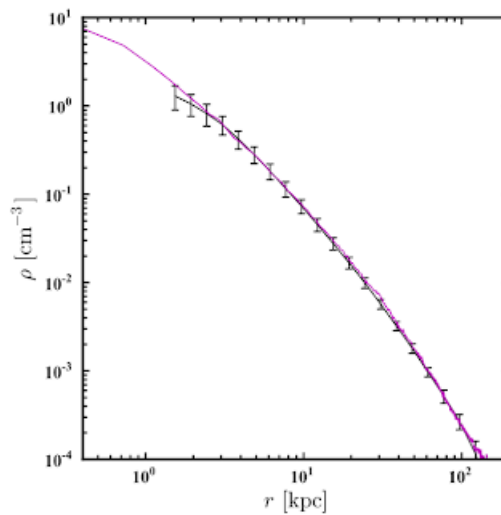
[note 2] M_{200} is defined as the mass enclosed in a sphere of mean density contrast of 200 with respect to the critical density; this follows the definition in Power et al. 2003 so it can be used in the estimation in the next sections (see [this link](#) for more details)

4. Analysis and Ideas for Paper "2"

- (1) **Obvious Ones:** assembly histories, accretion rates, merger trees, mass functions, low- and high-redshift comparison, etc. We can compare all the columns of the rockstar outputs. Also a direct comparison of positions and masses of the N most massive subhalos.
- (2) Studies on DM halo profile and and the mass resolution (raised by Mike Boylan-Kolchin)

It would be informative to comparing the density profile of the $10^{11} M_{\text{sun}}$ halo found in high-resolution zoom-in AGORA Proof-of-concept run with results from cosmological simulations with homogeneous mass resolution. This would demonstrate how representative the selected halo is in terms of a "fair" sample of similarly massive halos at large. It could also be used to highlight the importance of doing both DM and hydro zooms at this mass scale (and smaller): these halos are only resolved with 10^3 – 10^4 particles in state-of-the-art cosmological N-body simulations (Bolshoi / Millennium-II), so their structure and assembly histories are generally not very well resolved.

Attached is an example of such analysis: proof-of-concept run in RAMSES by Romain Teyssier (magenta) + a stacked profile for halos of the same mass selected from Millennium-II (black). The agreement is quite good, and residual differences are likely due to how substructure is treated.



- (3) Studies on DM halo mass function and the mass resolution (raised by Ken Nagamine)

O'Shea, et al. (2005) compared the Enzo and GADGET-2 results on dark matter halos in a cosmological box. In particular, they found that the AMR codes require larger computational resources for the root-grid to achieve the same DM halo mass function at low-mass end due to poorer force resolution at early times. Therefore it would be interesting to compare the number of low-mass halos (including substructures) in the zoom-in run, and see if and how this problem is alleviated by the AMR codes with a comparable computational resources. (For the information of the readers, all AMR codes in the Proof-of-concept DM-only run have used the same root-grid resolution.)

- (4) Studies of Velocity Distribution Function across code platforms (raised by Yao-yuan Mao, Risa Wechsler, Ji-hoon Kim)

Please see the papers: Mao et al. (astro-ph:1210.2721), Ling et al. (astro-ph:0909.2028), Kuhlen et al. (astro-ph:1308.1703)

5. Simulation Status

- (1) Working groups engaged: III (common cosmological ICs), IV (common analysis), VI (dwarf galaxies)
- (2) Participating codes and contact persons (led by Oliver Hahn):

Code	resp.	ICs work	Simulation run	Data transferred to NERSC	Halo Catalog?	Profiles?
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ART-I	Klypin	Y	to $z=0.1$			
ART-II	Feldmann					
CHANGA	Quinn	Y	to $z = 0$	Y		
ENZO	Kim, Simpson, Wise	Y				
GADGET-2/3	Choi, Hahn, Hobbs, Nagamine, Todoroki	Y	(G2, $z=0$, 160 snapshots)			
GASOLINE	Shen	Y				
PKDGRAV2	TBA	Y				
RAMSES	Hahn, Teyssier	Y				

6. Data Analysis Specifics/Deliverables

Data analysis scripts are stored in a [bitbucket mercurial repository](#).

- (1) Rockstar analysis of all 200 snapshots for all codes, run 'consistent trees' on these snapshots:
...
- (2) Compare sub-halo mass functions (in terms of $v_{\max}$, and in terms of v_{peak} , which requires the merger trees) for all codes at $z=0$, $z=1$, and high- z :
...
- (3) Compare density profiles for all codes at these redshifts:
...
- (4) Comparison of a selection of columns from the rockstar catalogs
...
- (5) a direct comparison of positions and masses of the N most massive subhalos, (shown overlayed on images as circles)
...
- (6) comparison of assembly histories, accretion rates
...
- (7) velocity distribution function, and some diagnostic of velocity structure at center
...

7. Results

to be filled soon